

CAD words, in plain English

Mechanical Engineering with CAD — makerspace club. Keep this beside you. Nothing here needs memorizing; look it up when a word turns up and carry on.

GETTING STARTED

CAD	Computer-aided design. Drawing a part on a computer accurately enough that it could actually be made.
Model	The part as it exists on the computer. Not a physical thing yet.
STL	A file format that holds a finished shape. It is what you save out when you want something printed.
Slicer	The program that turns your model into the moves a 3D printer follows. You open your STL in it just before printing.
Millimeter (mm)	The unit almost all of this uses. A US nickel is about 2mm thick, and a credit card just under 1mm. Always write the unit down.

DRAWING THE SHAPE

Sketch	A flat drawing of lines and circles. It is the starting point: you draw the outline first, then turn it into something solid.
Feature	The solid thing you make from a sketch. Pulling a rectangle up into a block is a feature.
Extrude	Pulling a flat shape straight up to give it thickness. A circle extruded becomes a cylinder.
Revolve	Spinning a flat outline around a line to make something round, like a knob or a pulley.
Dimension	A measurement written onto a drawing, fixing how big something is or where it sits.
Constrained (locked down)	When you have given a drawing enough measurements that nothing in it can wobble or slide. If a line still moves when you drag it, it is not locked down yet.
Feature tree	The list of every step you took, in order. You can go back and change an early step without redoing everything after it.
Workplane	A flat surface you choose to draw on. Useful when a feature needs to sit at an angle.
Pattern	Repeating one feature several times automatically, in a row, a ring, or mirrored across the part.

EDGES, WALLS AND CORNERS

Fillet	A rounded-off inside or outside corner. Rounded corners are stronger than sharp ones because stress spreads out instead of concentrating.
Chamfer	A corner cut off at an angle rather than rounded. Often used so a part slides into place more easily.
Gusset	A triangular web across an inside corner, bracing two faces so the joint does not bend.
Shell	Hollowing a solid part out, leaving walls of an even thickness.

Wall thickness

How thick the solid material is. Too thin and it snaps or fails to print; too thick and it wastes material and time.

FITTING PARTS TOGETHER

Assembly

Two or more parts put together, with rules about how they are allowed to move.

Mate (or joint)

The rule you set between two parts: fixed in place, hinged so it rotates, or sliding along a line.

Clearance

The deliberate gap between two parts so one can move inside the other. A rod 5mm across in a 5.4mm hole has 0.4mm of clearance.

Press fit

The opposite of clearance: the part is very slightly too big, so it grips when pushed in.

Tolerance

How far off the stated size a part is allowed to be and still work. Nothing is ever made exactly to size.

DRAWINGS FOR OTHER PEOPLE

Three-view drawing

The same object drawn from three directions, usually top, front and side, lined up so features match across them.

Datum

The edge or face that everything else is measured from. Pick one and measure from it consistently.

Title block

The box, usually bottom right, saying what the part is, who drew it, the scale and the units.

Extension line

The thin line running out from the part to where a measurement is written, so numbers sit outside the shape rather than on top of it.

MAKING IT REAL

Overhang

Part of a shape that sticks out over thin air. Steep overhangs print badly because there is nothing underneath holding them up.

Support

Temporary printed scaffolding under an overhang, snapped off afterwards.

Calipers

The measuring tool for reading a real object accurately. Use these rather than a ruler when the number matters.

If a word you need is not here, use the question box on any project and ask.